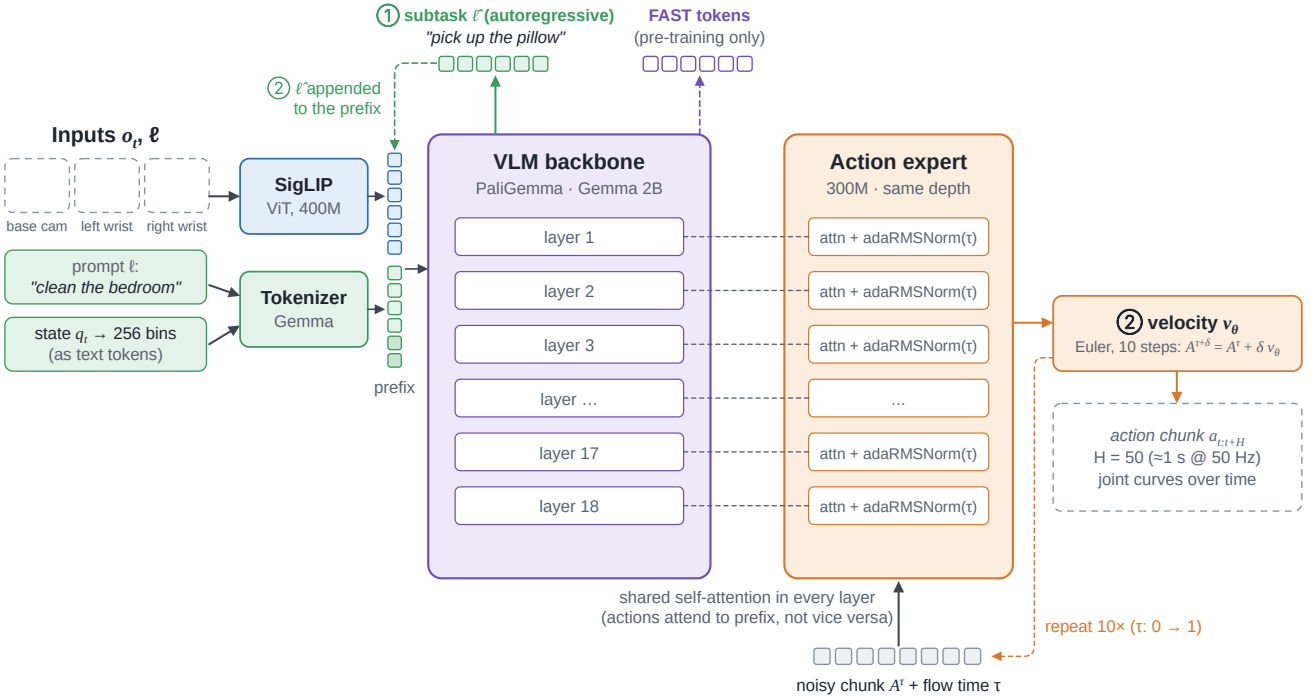


Attention mask (blockwise)

3×3 block matrix
[images+prompt+state | ℓ tokens | action tokens]

■ attend □ masked: prefix bidirectional, text causal

Training & inference

Pre-training (VLM only)
robot data → FAST tokens + subtask text
web data: captioning, VQA, detection
loss: next-token cross-entropy

Post-training (+ action expert)
flow matching on continuous chunks
$L = L_{CE} + \alpha\, L_{FM}$
inference: ① subtask → ② actions